SMART CART

Recipe recommendations and inventory-aware discounts

Offline presentation 12 November 2026

- Three demonstrations: original recommender -> synthetic workflow -> real reference-data integration.
- Goal: connect useful meal suggestions with transparent stock-aware promotion decisions.

THE PROBLEM

Two sides of the same shopping decision

Customer + seller

- Customers have selected items and need to know what else to buy.
- Sellers hold batches with different ages, expiry dates and excess-stock risk.
- The project links recipe coverage to useful published offers, purchase quantities and a readable invoice.

DEMO 1

Original recipe recommender

http://localhost:8601 main

- Normalize ingredient names and find recipes matching cart items.
- Show recommendation preferences and additional ingredient requirements.
- Baseline scope: no reliable budget pricing or seller checkout in this version.

DEMO 2

Controlled synthetic workflow

http://localhost:8602 small-test

- 15 recipes and 28 fictional ingredient entries, with synthetic prices and inventory.
- 20 prepared published offers; demonstrate customer cart -> combo -> simulated invoice.
- Scenario date: 5 November 2026. All market values and inventory examples are synthetic.

THE DISCOUNT POLICY

Transparent rules with margin protection

An interpretable heuristic, not a trained demand predictor

- Urgency combines age (35%), expiry (40%) and excess-stock risk (25%). Missing signals are renormalized.
- Corpus-frequency multipliers range from 1.0 to 1.3; recipe coverage is not sales demand.
- Maximum discount and minimum seller margin constrain quotes. Only published, valid offers reach checkout.

DEMO 3

Real reference data and complete workflow

http://localhost:8603 small-test-big-data

- 4,435 exactly linked price-and-calorie food codes; 75 fully component-priced USDA formulations.
- Historical USDA reference prices and nutrition are separate from simulated seller stock.
- Show Beans and brown rice: add missing component -> automatic offer -> whole packs -> invoice.
- Today’s prepared scenario: 7 simulated batches and 6 published offers.

CORRECT QUANTITIES AND PRICES

Recipe use differs from what must be purchased

A single consolidated row per product

- Recipe weights show one serving based on declared servings or an explicit portion estimate.
- Purchase quantities round up to full packs; the cart reports leftovers.
- Invoices combine combo and extra quantities, identifying discounted grams.
- Full purchased quantities are checked against usable tracked stock; catalog-only availability is unverified.

EVIDENCE AND LIMITATIONS

Demonstrate the result without overstating it

All three demo workflows verified with outbound connections blocked

- Original selection, synthetic combo checkout and real-data automatic discounts/invoices passed offline workflow checks.
- USDA formulations may contain cooked or packaged components; historical price estimates are not live grocery quotes.
- CPI uses a pinned September 2026 snapshot verified 8 November; seller prices remain separate.
- No real payment, stock deduction, measured waste reduction or validated real-world demand forecasting is claimed.

SUPERVISED ML EXPERIMENT

Learn the inventory risk score

Fractional logistic regression simulated sales outcomes

- Predict the fraction of stock unsold over up to seven days, using stock, recent sales, age, expiry and recipe frequency.
- 10,800 simulated examples; product-disjoint training, validation and test sets. Labels are simulated unsold outcomes, not copied rule scores.
- Held-out test MAE: 8.6 percentage points; constant-average baseline: 29.7 points. This measures simulation performance only.
- The ML score replaces manual priority weights. Discount caps and margin protection remain explicit business rules.

DISCUSSION

From reproducible prototype to retail integration

Thank you

- Next: verified current retail packs and prices, reviewed ingredient mappings and measured stock outcomes.
- The three local demos remain open. No internet or branch switching is needed.
- Recovery: RESET_DEMOS.sh -> START_DEMOS.sh -> reload tabs.